

MASHUK ELAHI

Electrical Engineer · Marine · Offshore · Energy · Semiconductor · Industrial

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PROFESSIONAL SUMMARY

Electrical Engineer with 16+ years delivering mission-critical LV/MV/HV power across marine, offshore, energy and semiconductor assets — FPSO topsides, jack-up rigs, offshore-wind substations, hybrid crew boats, CTVs, Ro-Pax ferries and 300 mm cleanrooms. Signature strength: first-pass class approvals across ABS-, BV-, LR- and DNV-classed projects and hybrid Li-ion BESS architectures, owning the full concept-to-handover lifecycle through ETAP modelling, construction, commissioning and the class interface.

Scale: 16+ years · FPSO to cleanrooms · crews of 100+. Certified Lead Auditor (ISO 9001 & 50001); building toward shore power, DC/LVDC zonal distribution and IACS E26/E27 OT-cyber for the decarbonised fleet.

CAREER HIGHLIGHTS

- **Up to ~50% lower fuel burn** — architected a 72 kWh Li-ion BESS hybrid power system on a 42 m Gen-4 Fast Crew Boat, saving up to ~50% fuel in low-load operation (first-pass class approval).
- **100% IEC 61850 conformity** — energised an offshore-wind substation topside to full protection-relay and alarm validation before sail-away.
- **Industry first** — delivered Malaysia's first solar–wind autonomous unmanned wellhead platform, eliminating routine diesel use and enabling remote operation.
- **Zero audit rework** — completed a 300 mm semiconductor cleanroom hook-up to 100% tool-up-time targets; prefabricated power skids cut on-site labour ~50%.

CORE COMPETENCIES

Electrical Power Systems LV/MV/HV switchgear · Power distribution · Protection coordination · Short-circuit, load-flow & arc-flash studies

Marine & Offshore FPSO topsides · Jack-up drilling rigs · Offshore-wind substations · CTVs · Ro-Pax ferries · Autonomous/electric tugs · Sea trials & sail-away

Hybrid & Renewable Li-ion BESS · BMS · DC UPS · DC/DC inverters · Shaft generators · Solar–wind hybrid · Shore power / cold ironing

Automation & Controls PLC · SCADA · IEC 61850 · PMS · IAS · DP1/DP2 · Integrated bridge & navigation · ABB 800xA Fire & Gas

E&I Compliance FAT · SAT · FMEA · Cause & effect · Hazardous-area (IECEx · ATEX · Zone 1/2) · Risk assessments · Class approval (ABS/DNV/LR/BV/NSCV)

Semiconductor & High-Tech 300 mm cleanrooms · UPS · Grounding · IEC 60204 · IEC 61511 · ISO 14644 · SS638 · SS650

Leadership Cross-functional coordination · Team leadership (100+) · Mentoring · Deliverable standardisation · KPI tracking · Continuous improvement

CORE TOOLS & SOFTWARE

ETAP · SKM PowerTools · MathCAD · MATLAB/Simulink · AutoCAD Electrical · ABB 800xA · Siemens · Yokogawa SCADA · Fluke DSX2-8000 · Python (engineering automation)

PROFESSIONAL EXPERIENCE

Electrical Engineer — Hybrid Marine Power

Jul 2024 – Present

Strategic Marine Pte Ltd · Marine / Newbuild · Singapore

Scope: Concept-to-handover LV power for hybrid newbuilds — 42 m Gen-4 Fast Crew Boats (100 PAX), an offshore-wind CTV series and passenger Ro-Pax ferries.

- Architected the complete LV power system for a 42 m Gen-4 Fast Crew Boat (3 × CAT C32; 3 × 150 ekW; VEEM propulsion) — SLDs, load lists, cable schedules — first-pass class approval (programme portfolio spans ABS, BV, LR & DNV).

- Integrated a 72 kWh Li-ion BESS, BMS, DC/DC inverters and a 90 kW shaft generator into a hybrid drivetrain under LR Hybrid Power notation — cut fuel burn by up to ~50% in low-load operation.
- Delivered electrical design across a multi-hull Gen-4 programme and offshore-wind CTV series, plus a 58 m Ro-Pax and 43 m fast ferry — dual 380/220 V AC + 24 V DC, interlocked shore power; BV / LR / AMSA-NSCV.
- FAT-validated ABB azimuth propulsion / PMS on an autonomous-electric tug; led cross-fleet cathodic-protection & stray-current engineering; standardised departmental deliverables and KPI tracking.

Senior Project Engineer — Semiconductor & High-Tech Facilities

Feb 2023 – Jul 2024

Exyte · Semiconductor / Cleanroom · Singapore

Scope: 300 mm semiconductor cleanroom power hook-up, functional safety and ISO 50001 energy stewardship.

- Led 300 mm cleanroom power hook-up to IEC 60204 / 61511 / ISO 14644 — zero audit rework against 100% tool-uptime targets.
- Introduced prefabricated power skids — cut on-site labour ~50% and removed contamination risk.
- Validated PLC/SCADA interlocks and functional safety to SS638, SS650 and IEC 62061 — cleared audit first pass.
- Drove ISO 50001 energy stewardship via smart load monitoring — lowered fab energy intensity.

Electrical & Instrumentation Engineer — Hazardous-Area, Renewables & Power

Jan 2021 – Jan 2023

Phoenix Mecano (S.E.A.) Pte Ltd · Industrial / Offshore · Singapore

Scope: Hazardous-area E&I, autonomous renewables and onshore power generation.

- Pioneered Malaysia's first solar-wind autonomous unmanned wellhead platform (24 V DC solar/battery bus) — eliminated routine diesel use.
- Directed an onshore cogeneration plant E&I scope — transformers, HV/LV switchgear, 400 kVA UPS, tie-in and instrument-loop design; ABB / Siemens / Yokogawa SCADA.
- Designed IECEx/ATEX Zone 1/2 hazardous-area systems — Ex equipment selection, IS barriers and zoning studies.
- Standardised compliance documentation as certified Lead Auditor (ISO 9001 & ISO 50001).

Electrical & Instrumentation Superintendent — FPSO, Rigs & Offshore Wind

Nov 2012 – Dec 2020

Keppel FELS (Keppel Offshore & Marine) · Offshore / Marine · Singapore

Scope: FPSO topsides, ultra-deepwater rigs, semisubmersibles and an offshore-wind substation — crews up to 100+.

- Superintended teams up to 100+ across fabrication, installation and integrated commissioning of FPSO topsides — energised turret, HV switchgear and ABB 800xA Fire & Gas.
- Energised an offshore-wind substation topside to 100% IEC 61850 conformity before sail-away.
- Commissioned ultra-deepwater jack-up drilling rigs and semisubmersible accommodation modules through FMEA and sea trials.
- Delivered integrated bridge & navigation systems (gyro, VDR, alarm & watch); authored 30+ E&I risk assessments; mentored 15+ engineers via Innovation Quality Circles.

Assistant Electrical Engineer

Sep 2009 – Nov 2012

Ananda Shipyard & Slipways Ltd · Newbuild / Marine · Bangladesh

- Led E&I crews of 15–30 on vessel and rig newbuilds — commissioned azimuth thrusters, controllable-pitch propellers and bridge navigation to ABS class; cable schedules and voltage-drop checks against milestone sign-off.

EDUCATION

Master's in Sustainable Electricity Generation (60 ECTS) — Universidad Católica de Murcia (UCAM), Spain

Apr 2026

MSc Digital Transformation — London Metropolitan University, UK

Apr 2026

MBA — University of Sunderland, UK

Jun 2025

BSc Electrical & Electronic Engineering — Bangladesh University

2008

Diploma in Electronics Engineering — Jessore Polytechnic Institute

Jan 2004

CERTIFICATIONS & TRAINING

- **Certified Lead Auditor** — ISO 9001:2015 Quality Management (Jul 2023) & ISO 50001:2018 Energy Management (GAQM, Feb 2023).
- **IECEX Ex 001** — Basic Principles of Protection in Explosive Atmospheres (valid to Oct 2027).
- **Safety:** SSSC · WSQ WSH Management in Construction · SEC(K) Multi-skilling Construction Personnel (Jan 2024) · Work at Height (Supervisor) · Confined Space · WSH in Process Plant.
- **Technical:** Roxtec System Installer · CMP Cable Gland Installer · RAYCHEM APKT/EPKT · Fluke DSX2-8000 Copper Certifier · ABB KFELS B-Class Jack-up Rig LV Switchboard Training.
- **Emerging & future-ready (in progress, 2025–2026):** IACS UR E26/E27 OT cybersecurity · IEC/ISO/IEEE 80005-1 shore power · IEC 61660 DC microgrids · IEC 62619 battery safety · CompEx Ex01–Ex04.
- **Published thought leadership:** four LinkedIn technical articles — IACS E26/E27, 80005-1 shore power, IEC 61660 DC microgrids and IEC 62619 battery safety.